

MINOR PROJECT · CS/DS

Scheme-AI

Government Scheme Eligibility &
Application AI Assistant

Agentic AI · RAG · NLP · LLM



*Submitted by: Bhavna Sharma (0827CD231018),
Faiz Hashmi (0827CD231030)
Lavina fand (0827CD231046),
Sameer Jaiswal (0827CD231061)*

*Submitted To: Prof. Mayank Bhatt
Prof. Mahendra Verma*

Crores of Eligible Citizens Miss Out on Benefits

53%

of India's salaried workforce
lacks any social security benefit

1,700+

central & state govt schemes
in active operation

~40%

of welfare budgets remain
unspent or unutilised

ROOT CAUSES



No Awareness

Schemes exist on paper;
citizens never hear of them



Complex Forms

Government jargon is
hard to decode



Scattered Info

No single place to check
all eligible schemes



Language Gap

Most portals English-only;
no regional language support

GovAI: An Agentic AI That Does the Heavy Lifting

01



Conversational Profiling

AI asks simple questions
(age, income, state, occupation)

02



Smart Eligibility Matching

RAG retrieves & ranks
relevant schemes

03



Plain Language Explanation

Benefits explained simply
in user's language

04



Step-by-Step Application Guide

Documents, portals,
deadlines — all guided

What makes it different from myScheme.gov.in?

✓ Conversational, not form-based

✓ Multilingual & voice-ready

✓ Agentic — guides end-to-end

✓ Explains WHY you qualify

Who Benefits & How



Farmer

Kisan from Bihar

Eligible for:

PM-KISAN, PMFBY (crop insurance), Soil Health Card



Student

Girl from tribal area

Eligible for:

National Scholarship Portal, Beti Bachao, ST/SC grants



Migrant Worker

Construction labourer, Delhi

Eligible for:

e-Shram, BOCW welfare, PMAY (housing)



Small Biz Owner

Women entrepreneur

Eligible for:

Mudra Yojana, Stand-Up India, Startup India



Senior Citizen

Retired farmer, 65+

Eligible for:

PMVVY, Atal Pension Yojana, Ayushman Bharat



Pregnant Woman

First-time mother

Eligible for:

PMMVY (maternity benefit), JSY, Poshan Abhiyaan

How It's Built

User Interface	Streamlit · WhatsApp API
Agent Layer	LangChain Agents · LangGraph · Conversation Memory
RAG Pipeline	Document Loader → Text Splitter → Embeddings → ChromaDB → Retriever
LLM Core	GPT-4 · Prompt Engineering · Output Parser
Knowledge Base	Scraped myScheme.gov.in · Scheme PDFs · State portals · Auto-updated pipeline

KEY TOOLS

 LangChain	Agent orchestration
 LangGraph	Stateful multi-step flows
 OpenAI / Groq	LLM reasoning engine
 ChromaDB	Vector storage & retrieval
 Bhashini API	Indian language support
 FastAPI	Backend REST API
 Streamlit	Rapid UI prototyping

Honest Challenges We Must Address

Data Freshness

HIGH

Scheme eligibility rules change frequently. Static knowledge bases go stale, leading to incorrect guidance.

LLM Hallucination

HIGH

LLMs can confidently generate wrong eligibility details, especially for obscure state-level schemes.

Digital Divide

HIGH

Most rural beneficiaries lack smartphones or internet — the very people who need help most.

Language Coverage

MED

India has 22 scheduled languages + dialects. Full NLP support for all is complex and resource-heavy.

Data Privacy

MED

Collecting sensitive profile data (income, caste, Aadhaar-linked info) creates privacy and compliance risk.

Coverage Gaps

MED

Not all state/district-level schemes are digitally documented; AI cannot guide what it doesn't know.

Mitigation Strategy

Data Freshness	Automated scraping pipeline on myScheme.gov.in + webhook alerts on scheme updates. Data versioned in vector DB.	Scrapy · Apify Pinecone metadata
Hallucination	Strict RAG grounding — AI only answers from retrieved context. Confidence scoring; unknown queries flag to human.	RAGAS eval Guardrails AI
Digital Divide	WhatsApp-first deployment (800M+ Indian users). IVR voice-call fallback for feature-phone users via Bhashini.	Twilio · Bhashini WhatsApp Business API
Language Coverage	Integrate Bhashini (Govt of India's AI stack) covering 22 scheduled languages with TTS/STT support.	Bhashini · IndicTrans2 AI4Bharat models
Data Privacy	No PII stored permanently. Aadhaar lookup via DigiLocker API only. Session data encrypted, deleted after 24h.	DigiLocker API DPDP Act 2023

THE OPPORTUNITY

**If even 10% more eligible citizens
find and apply for their schemes —**
that's millions of lives materially changed.

Real-world Impact

Technically Feasible

Uniquely Indian Problem

Scalable with AI

GovAI — Built for Bharat

Agentic AI · RAG · LangChain · LangGraph · Bhashini · FastAPI · Streamlit